1.1 What is a digital twin?

In his 1991 book *Mirror Worlds*, computer scientist David Gelertner imagines building a physical model of a city on your living room floor containing miniature models of buildings and cars, combined with blackboards radiating information related to other things people care about, like the city budget, the air quality, and the waiting time in the emergency room. An army of people would gather data in the city and pass it back to your living room to update this model, allowing you to see, at a glance from your sofa, what is going on throughout the city and use this to influence your own decisions. He then imagines this model living in software, allowing countless people to observe the city simultaneously, each at their preferred level of detail.

Around the same time, Michael Grieves was developing similar concepts in the manufacturing world. In 2002, while teaching at the University of Michigan, Grieves presented what he called the "Mirrored Spaces Model" and later the "Information Mirroring Model"—a framework for creating virtual representations of physical products throughout their lifecycle. His work laid the groundwork for what would eventually be called digital twins in industrial settings.

What Gelertner and Grieves envisioned has become reality with the emergence of digital twins. The term has been used in many contexts since then and can be defined in different ways, but for this book, a digital twin is defined as:

> *A digital representation of a physical system that is updated with real-world data. It helps users understand current conditions, monitor performance, and simulate scenarios to make better decisions based on clearly defined objectives. Digital twins can also act on the physical system, sending instructions to adjust or optimize its behavior.*

This definition emphasizes that digital twins should not be built merely because the technology is available, but should serve specific business, operational, or research goals. By providing a computer-based model that mirrors something physical in reality, they help users to comprehend both